கல்வி வானம்

- கல்வி வானம் என்பது தமிழ்மூலம் இலவசமாக கல்வி கற்பிக்கும் ஒரு தளம்.
- இங்கு பாடங்கள் எளிமையாகவும் தெளிவாகவும் வழங்கப்படுகின்றன.
- மாணவர்களின் சந்தேகங்கள் இங்கே உடனடியாக தீர்க்கப்படுகின்றன.
- கற்றல் எல்லோருக்கும் எளிதாக கிடைக்க வேண்டும் என்பதே கல்வி வானத்தின் நோக்கம்.

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CPP - 1 SCOPE OF VARIABLES

```
#include<iostream>
using namespace std;

int x=5;

int main()
{
   int x=6;
   {
      int x =10;
      cout << "\n local scope variable x = "<< x;
   }
   cout << "\n function scope variable x =" <<x;

   cout << "\n global scope variable x =" << ::x;
   return 0;
}

//OUTPUT
Local scope variable x=10
Function scope variable x=6
global scope variable x=5
```

CPP - 2 FACTORIAL PROGRAM (Using Recursive function)

```
#include<iostream>
using namespace std;

int fact (int n)
{
   if(n== 0 || n == 1)

      return 1;
   return n*fact(n-1);
}

int main()
{
int n;
   cout << "\n Enter n value... " << endl;
   cin >> n;
   cout << "\n Factorial of "<<n<<" = "<<fact(n);
   return 0;
}

//OUTPUT
Enter n value... 5

Factorial of 5 = 120
```

CPP - 3 ODD OR EVEN NUMBER (Using if..... else)

```
#include<iostream>
using namespace std;

void oddeven(int n)
{
   if(n % 2 == 0)
      cout << "\n The given number is even";
   else
      cout << "\n The given number is odd";
}

int main()
{
   int a;
   cout << "\n Enter a number ... ";
   cin >> a;

   oddeven(a);

   return 0;
}
```

```
//OUTPUT
Enter a number ... 5

 The given number is odd
```

CPP - 4 PERCENTAGE OF MARKS AND DIVISION (Using Switch Case)

```
#include<iostream>
using namespace std;

int main()
{
   float percent;
   cout << "\n Enter your percentage... ";
   cin >> percent;
   cout << "\n You scored " << percent << "%";
   int n = percent / 10;

   switch(n)
   {
   case 10:
   case 9:
   case 8:
      cout << "\n You have passed with Distinction";
      break;
   case 7:
   case 6:
      cout << "\n You have passed with First Division";
      break;
   case 5:
   case 4:
      cout << "\n You have passed with Second Division";
      break;
   case 3:
      cout << "\n You have passed with Third Division";
      break;
   default:
      cout << "\n Sorry: You have failed" << endl;
   }
   return 0;
}
```

```
//OUTPUT
Enter your percentage... 75
You scored 75%
You have passed with First Division
```

CPP -5 PALINDROME (REVERSE OF NUMBER)

```
#include<iostream>
using namespace std;

int main()
{
   int n,num,digit,rev=0;

   cout << "Enter a positive number ";
   cin >>num;
   n=num;
   while(num)
   {
      digit=num%10;
      rev=(rev*10)+digit;
      num=num/10;
   }
   cout << "The reverse of the number is " <<rev<<endl;
   if(n==rev)
      cout << "The number is a palindrome \n";

   else
      cout << "The number is not a palindrome \n" << endl;

   return 0;
}
//OUTPUT
Enter a positive number 1221
The reverse of the number is 1221
The number is a palindrome
```

CPP - 6 SUM OF NATURAL NUMBER (Using FOR LOOP)

```
#include<iostream>
using namespace std;

int main()
{
   int sum = 0, i;

   for(i = 1; i <= 10; i++)
   {
      cout << "\n" << i;
      sum += i;
   }

   cout << "\n Sum of the series = " << sum;
   return 0;
}

//OUTPUT
1
2
3
4
5
6
7
8
9
10
```

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```
Sum of the series = 55
```

CPP - 7 CALL BY VALUE

```
#include<iostream>
using namespace std;

void swap(int n)
{
   n = 20;
}

int main()
{
   int m = 10;

   cout << "\n Value of m before calling swap function....." << m;
   swap(m);
   cout << "\n Value of m after calling swap function....." << m;

   return 0;

}

//OUTPUT
Value of m before calling swap function.....10
Value of m after calling swap function.....10
```

CPP - 8 CALL BY REFERENCE

```
#include<iostream>
using namespace std;

void swap(int & n)
{
   n = 20;
}

int main()
{
   int m = 10;

   cout << "\n Value of m before calling swap function....." << m;
   swap(m);
   cout << "\n Value of m after calling swap function....." << m;

   return 0;

}

//OUTPUT
Value of m before calling swap function.....10
Value of m after calling swap function.....20
```

CPP - 9 FUNCTION OVERLOADING

```
#include<iostream>
using namespace std;
void print(int i)
{
   cout << "\n integer i =" <<i;
}
void print(double f)
{
   cout << "\n double f =" << f;
}
void print(char * c)
{
   cout << "\n char c =" <<c;
}
int main()
{
   print(10);
   print(10.01);
   print("Ten");
   return 0;
}

//OUTPUT
integer i =10
double f =10.01
char c =Ten
```

CPP - 10 OPERATOR OVERLOADING ((-) minus operator is overloading)

```
#include<iostream>
using namespace std;

class negative
{
   int n;
public:
   void accept()
   {
      cout << "\n Enter a number....";
      cin >> n;
   }

   void operator-()
   {
      n = -n;
   }

   void disp()
   {
      cout << "\n after overloading the minus operator n = " <<n;
   }

};

int main()
{
   negative m;
   m.accept();
   -m;
   m.disp();
   return 0;
}
```

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```
// OUTPUT
Enter a number....5
after overloading the minus operator n = -5
```

CPP - 11 CLASS AND CONSTRUCTOR (EXAMPLE)

```
#include<iostream>
using namespace std;

class student
{
int m1, m2, total;
public:
student()
{
total = 0;
}

void getdata()
{
   cout << "\n Enter marks in two subjects..." << endl;
   cin >> m1 >> m2;
   total = m1 + m2;
}

void printdata()
{
   cout << "\n Total marks = " << total << endl;
}

};

int main()
{
student stud1;
stud1.getdata();
stud1.printdata();
return 0;
}
```

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```
//OUTPUT
Enter marks in two subjects...
20
30
Total marks = 50
```

CPP-12 BREAK AND CONTINUE (EXAMPLE)

```
#include<iostream>
using namespace std;

int main()
{
   for (int i=1; i<=10; i++)
   {

      if (i==3)

         continue;

      cout<<"\n i value="<<i;

      if (i==6)

         break;
   }
}
//OUTPUT
 i value=1
 i value=2
 i value=4
 i value=5
 i value=6
```

CPP-13 STRUCTURE(EXAMPLE)

```
#include<iostream>
using namespace std;
struct student
{
   char name[15]    ;
   int age;
   float ht;
} stud1;
int main()
{
   cout << "\n Enter the name.... ";
   cin>>stud1.name;
   cout << "\n Enter the age.... ";
   cin >> stud1.age;
   cout << "\n Enter the height...";
   cin >> stud1.ht;
   cout << "\n name , age and height of stud1 = " <<
stud1.name<<"\t"<<stud1.age << "\t" << stud1.ht;
   return 0;
}
//OUTPUT
Enter the name.... raja
 Enter the age.... 15
 Enter the height...45
 name , age and height of stud1 = raja  15   45
```

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CPP-14 MATRIC ADDITION (TWO DIMENSIONAL ARRAY)

```
#include <iostream>
using namespace std;

int main()
{
   int i, j, r, c, m1[10][10], m2[10][10], sum[10][10];

   cout << "\n Enter the row size of the array: ";
   cin >> r;

   cout << "\n Enter the col size of the array: ";
   cin >> c;

   cout << "\n Enter the elements of the first matrix:\n";
   for(i = 0; i < r; i++)
   {
      for(j = 0; j < c; j++)
      {
         cin >> m1[i][j];
      }
   }

   cout << "\n Enter the elements of the second matrix:\n";
   for(i = 0; i < r; i++)
   {
      for(j = 0; j < c; j++)
      {
         cin >> m2[i ][j];
      }
   }

   cout << "\n Sum of the two matrices:\n";
   for(i = 0; i < r; i++)
   {
      for(j = 0; j < c; j++)
      {
         sum[i][j] = m1[i][j] + m2[i][j];
         cout << sum[i][j] << "\t";
      }
      cout << "\n";
   }

   return 0;
}
```

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```
//OUTPUT
Enter the row size of the array: 2
 Enter the col size of the array: 2
 Enter the elements of the first matrix:
1
1
1
1

 Enter the elements of the second matrix:
1
1
1
1

 Sum of the two matrices:
2     2
2     2
```

All the Best For get Good Marks On Your Exam

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knowledge has no limit